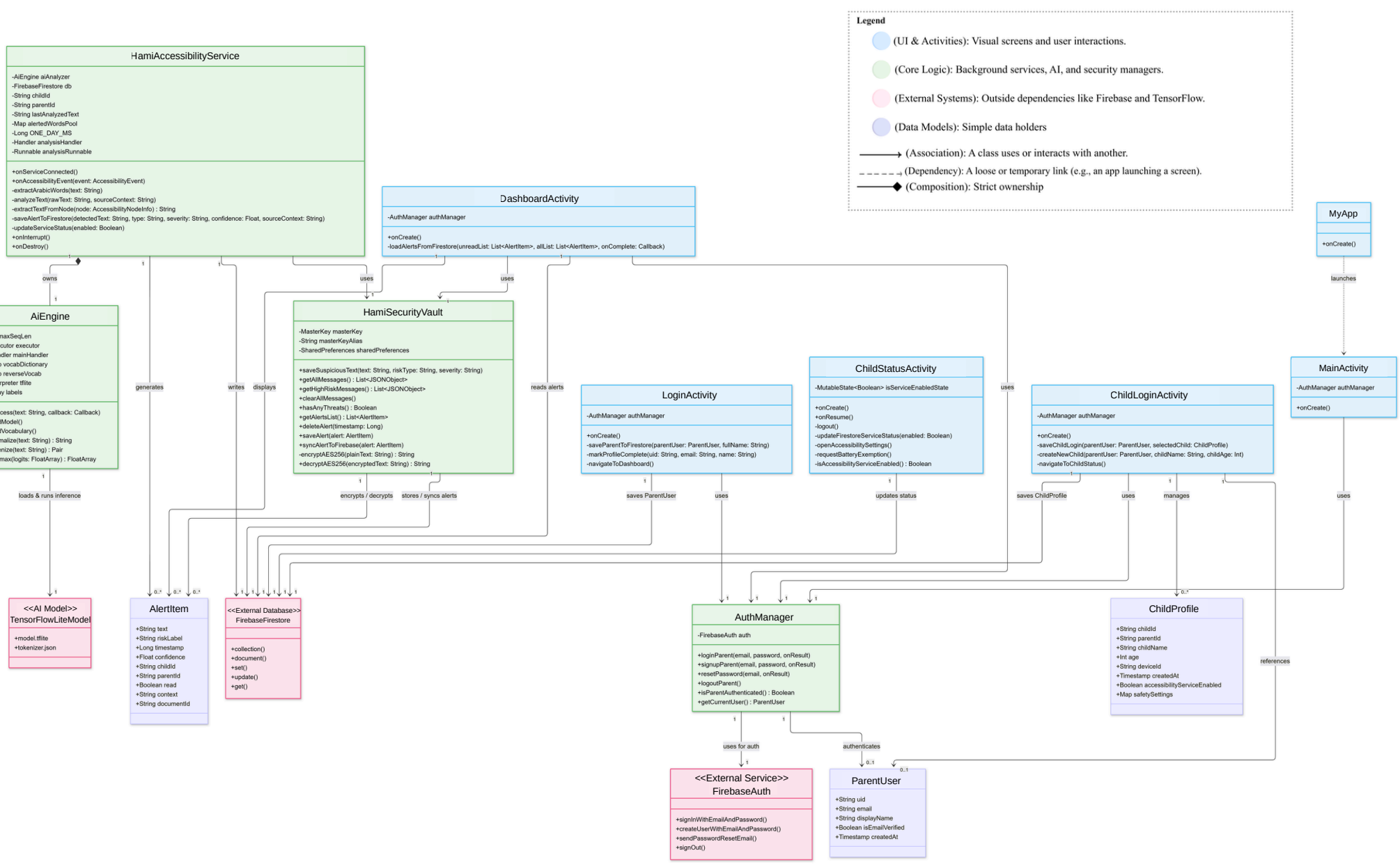

Legend
(UI & Activities): Visual screens and user interactions.
(Core Logic): Background services, AI, and security managers.
(External Systems): Outside dependencies like Firebase and TensorFlow.
(Data Models): Simple data holders
(Association): A class uses or interacts with another.
(Dependency): A loose or temporary link (e.g., an app launching a screen).
(Composition): Strict ownership
HamiAccessibilityService
-AiEngine aiAnalyzer
-FirebaseFirestore db
-String childId
-String parentId
-String lastAnalyzedText
-Map alertedWordsPool
-Long ONE_DAY_MS
-Handler analysisHandler
-Runnable analysisRunnable
+onServiceConnected()
+onAccessibilityEvent(event: AccessibilityEvent)
-extractArabicWords(text: String)
-analyzeText(rawText: String, sourceContext: String)
-extractTextFromNode(node: AccessibilityNodeInfo) : String
-saveAlertToFirestore(detectedText: String, type: String, severity: String, confidence: Float, sourceContext: String)
-updateServiceStatus(enabled: Boolean)
+onInterrupt()
+onDestroy()
DashboardActivity
-AuthManager authManager
+onCreate()
-loadAlertsFromFirestore(unreadList: List<AlertItem>, allList: List<AlertItem>, onComplete: Callback)
MyApp
+onCreate()
launches
owns
uses
uses
AiEngine
-Int maxSeqLen
-Executor executor
-Handler mainHandler
-Map vocabDictionary
-Map reverseVocab
-Interpreter tflite
-Array labels
+process(text: String, callback: Callback)
-loadModel()
-loadVocabulary()
-normalize(text: String) : String
-tokenize(text: String) : Pair
-softmax(logits: FloatArray) : FloatArray
HamiSecurityVault
-MasterKey masterKey
-String masterKeyAlias
-SharedPreferences sharedPreferences
+saveSuspiciousText(text: String, riskType: String, severity: String)
+getAllMessages() : List<JSONObject>
+getHighRiskMessages() : List<JSONObject>
+clearAllMessages()
+hasAnyThreats() : Boolean
+getAlertsList() : List<AlertItem>
+deleteAlert(timestamp: Long)
+saveAlert(alert: AlertItem)
+syncAlertToFirebase(alert: AlertItem)
-encryptAES256(plainText: String) : String
+decryptAES256(encryptedText: String) : String
LoginActivity
-AuthManager authManager
+onCreate()
-saveParentToFirestore(parentUser: ParentUser, fullName: String)
-markProfileComplete(uid: String, email: String, name: String)
-navigateToDashboard()
ChildStatusActivity
-MutableState<Boolean> isServiceEnabledState
+onCreate()
+onResume()
-logout()
-updateFirestoreServiceStatus(enabled: Boolean)
-openAccessibilitySettings()
-requestBatteryExemption()
-isAccessibilityServiceEnabled() : Boolean
ChildLoginActivity
-AuthManager authManager
+onCreate()
-saveChildLogin(parentUser: ParentUser, selectedChild: ChildProfile)
-createNewChild(parentUser: ParentUser, childName: String, childAge: Int)
-navigateToChildStatus()
MainActivity
-AuthManager authManager
+onCreate()
generates
writes
displays
reads alerts
uses
loads & runs inference
encrypts / decrypts
stores / syncs alerts
saves ParentUser
uses
updates status
saves ChildProfile
uses
manages
uses
<<AI Model>>
TensorFlowLiteModel
+model.tflite
+tokenizer.json
AlertItem
+String text
+String riskLabel
+Long timestamp
+Float confidence
+String childId
+String parentId
+Boolean read
+String context
+String documentId
<<External Database>>
FirebaseFirestore
+collection()
+document()
+set()
+update()
+get()
AuthManager
-FirebaseAuth auth
+loginParent(email, password, onResult)
+signupParent(email, password, onResult)
+resetPassword(email, onResult)
+logoutParent()
+isParentAuthenticated() : Boolean
+getCurrentUser() : ParentUser
ChildProfile
+String childId
+String parentId
+String childName
+Int age
+String deviceId
+Timestamp createdAt
+Boolean accessibilityServiceEnabled
+Map safetySettings
references
uses for auth
authenticates
<<External Service>>
FirebaseAuth
+signInWithEmailAndPassword()
+createUserWithEmailAndPassword()
+sendPasswordResetEmail()
+signOut()
ParentUser
+String uid
+String email
+String displayName
+Boolean isEmailVerified
+Timestamp createdAt